Stable DNA Storage Encoding Scheme Based on Repeating Substring Tree

Jieqiong Wu, Penghao Wang, Yanfen Zheng, Bin Wang, *Member, IEEE*, Qiang Zhang, and Pan Zheng, *Senior Member, IEEE*

***Abstract*—DNA storage is considered to be a promising storage media in the current era of data explosion. DNA encoding is the beginning of the DNA storage process and lays the foundation for subsequent processes. However, many encoding methods suffer from low encoding rate, do not satisfy important constraints, or have insufficient sequence stability. To address these issues and improved sequences stability, this paper proposes a novel approach called the Repeating Substring Tree Encoding (RSTE) method. The method begins by applying the Longest Substring Backtracking Method (LSBM) to identify the longest repeated substrings within the binary file. These substrings are then encoded into compact DNA motifs using Huffman encoding. In contrast to the ideal coding density of 2 bits per nucleotide (2 bit/nt) targeted by previous studies, RSTE enhances the encoding rate by 13% through efficient utilization of repeated substrings. Furthermore, the DNA sequences generated by the RSTE method successfully meet three biological constraints: run-length limitation, GC content balance and end constraints. The experimental results of minimum free energy and melting temperature indicate that the stability of the sequences encoded by RSTE is also greatly improved. A series of experiments showed that the sequences encoded by RSTE have a higher coding rate, satisfy constraints, and are more stable.**

***Index Terms*—DNA storage, DNA encoding, RSTE, lossless encoding.**

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I. Introduction

With an ever-increasing amount of data, DNA storage has received extensive at-tention due to its extremely high storage density and extremely long storage time [1]. The main idea of DNA data storage is to map digital binary data to DNA nucleotide se-quences according to certain conditions. DNA sequences can be stored and read by high-throughput sequencing (HTS) and converted back into binary information [2], [3]. In recent years, the technology of DNA storage has been continuously developed [4], [5]. Church et al. [2] encoded information using next-generation DNA synthesis and sequencing technologies in 2012. Although this method has a coding rate of only 1 bits/nt, it reduces the number of repeated bases. Yazdi et al. [6] proposed a storage architecture with high storage efficiency in 2015. The following year, they applied an integrated processing pipeline to encode data [7]. In 2020, Tomek et al. [8] achieved file storage through chemical processing and improving the scalability of data storage system. Lee et al. [9] proposed a multi-enzyme DNA synthesis method that uses maskless photoli-thography.

DNA storage involves six steps: encoding, synthesis, preservation, retrieval, se-quencing, and decoding [10]. Encoding refers to converting the data to be stored into a base sequence according to a certain encoding method. Much work has been done to improve the encoding rate [11]. Goldman et al. [12] used Huffman encoding for the first time in DNA storage in 2013, which effectively increased the encoding potential to 1.58 bits/nucleotide (nt). However, the encoding method caused slow sequencing. In 2015, Grass et al. [13] proposed error-correcting codes for correcting errors, resulting in an encoding density of 1.14 bits/nt. Bornholt et al. [14] proposed XOR encoding method in 2016, which has a density more than twice that of Huffman (0.88 bits/nt). To further increase the encoding density, Hwang et al. [15] used degenerate base pairs to achieve high information density. In 2016, Blawat et al. [16] proposed a meth-od named forward error correction encoding with data density reaching 0.92 bits/nt. Kiah et al. [17] introduced DNA storage channels and proposed new asymmetric en-coding techniques through modeling to counter the effects of synthesis and sequence noise. A file with a capacity of hundreds of megabytes may require hundreds or even thousands of DNA sequences, and a DNA pool may store tens to hundreds of different files. Therefore, to improve the storage efficiency, appropriate redundancy is added to the payload, and there are two basic types of redundancy: physical redundancy and logical redundancy [18]. Physical redundancy is used to store copies of DNA sequences, and logical redundancy

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is other information outside the stored sequence, such as the primer, address bits, and parity bits. The higher the redundancy, the more error toler-ance there exists. The upper limit of redundancy is the maximum data that can be stored per nucleotide [19]. In 2017, Erlich et al. [20] proposed an encoding method for fountain codes, which raised the encoding potential to a high value of 1.98 bits/nt. Yazdi et al. [7] designed an integrated processing pipeline, and they exploited efficient. ortable sorting with new iterative calibration and deletion error-correcting code. In 2018, Song et al. [21] proposed an encoding method that satisfies the run-length limit (RLL) constraint and GC content constraint. Limbachiya et al. [22] proposed an algorithm that generates DNA sequences with a minimum specified Hamming distance between each sequence, such that all codewords follow the RLL constraint and GC-content constraint. In 2019, Wang et al. [23] devised a DNA data storage scheme with variable-length bases. The experimental results showed that the scheme achieved a net information density of 1.67 bits/nt and 91% information capacity. Choi [24] and Anavy [25] introduced additional degenerate base characters on the traditional four bases, which improved the encoding rate. In 2022, Ping et al. [26] proposed Yin-Yang Coding, a DNA encoding algorithm that reduces long homopolymers and controls GC content. Verified by in vitro and in vivo tests, YYC achieved an average recovery rate of 99.9%, a minimum of 87.53%, and storage density near the theoretical maximum.In 2023,Welzel et al. [27] proposed DNA-Aeon, a concatenated arithmetic coding scheme for DNA storage supporting user-defined constraints like GC content, homopolymer length, and sequence pattern avoidance.

Due to the physical and biological properties of DNA sequences, the sequences are prone to secondary structures and other errors during the storage process of DNA. The cost of DNA storage mainly depends on the cost of writing data through DNA synthesis and the cost of reading data through DNA sequencing [28]. The cost of writing data is the dominant component [29]. DNA encoding is the beginning of the DNA storage process, it is necessary to encode high-quality storage sequences to avoid errors in subsequent steps, and simultaneously, it is necessary to increase the encoding rate to control storage cost [30]. Good encoding can lay a good foundation for subsequent work. If the encoding work is performed well, subsequent procedures can also be completed relatively smoothly. Therefore, it is particularly important to develop a new, high-quality encoding method. However, DNA sequences are different from prevailing silicon-based storage systems, given the former's special physical and biological properties and other errors, such as non-specific hybridization, fragmentation, and secondary structures, that can occur [31]. Therefore, in the encoding process, on the one hand, it is necessary to increase the encoding rate to save storage cost, and on the other hand, to encode high-quality storage sequences to avoid errors in subsequent storage steps also important. In the process of binary data conversion to a base sequence, the most commonly used methods are as follows: the first is the mapping template; that is, A, T, C, and G are used to represent 00, 01, 10, and 11, respectively. The second encoding method is to divide the source code into data blocks of equal length according to a fixed value and fill in the insufficient ones with zeros. Then, one finds the repetition frequency of each data block and performs the encoding operation ac-cording to different data blocks and their frequencies. The first encoding method can reach the theoretical maximum encoding value of 2, but the stability of the encoded sequence is poor and does not meet the physical and biological constraints that need to be considered for DNA storage sequences. It is difficult for the second encoding meth-od to reach the theoretical maximum encoding value, and it may cause data corruption. This paper proposes a new encoding method named the Repeating Substring Tree Encoding (RSTE) Method. The encoding method has the following steps: 1) input the source data to be encoded; 2) find the repeated substrings in the source data through the Longest Substring Backtracking Method (LSBM); 3) generate the hash table of cor-responding substrings and positions; 4) according to the hash table, build a Huffman tree to generate the encoding result of the corresponding substring; and 5) generate the final encoding result according to the encoding result of the substring. Compared with other encoding methods, this method increases the encoding rate and realizes lossless compression of data. In addition to focusing on the coding rate that many researchers have focused on, as a new encoding method, RSTE also considers the RLL constraint, GC-content constraint, and End-constraint that the stored sequences should satisfy. The sequence encoded by our proposed encoding method improves the stability of the encoded sequence and makes the encoded sequence more suitable for the real DNA storage process.

The structure of the rest of this paper is as follows. Section II provides a detailed description of the DNA storage sequence structure and RSTE encoding process. Section III explains the biological constraints satisfied by the encoded sequence. Section IV discusses the coding efficiency and coding quality and compares this work with other achievements in each aspect. Finally, we summarize this work and identify avenues for future work.

II. Encoding Method and Constraints

Encoding is the first step in the DNA storage process. It is necessary not only to improve the encoding rate to reduce the cost of synthesis and sequencing but also to consider the stability of the sequence used for storage. In this chapter, we propose the RSTE method, which uses LSBM to search repeat substrings, uses a Huffman tree to construct the DNA storage coding set, and satisfies important constraints.

A. The Structure of DNA Storage Sequence

Sequences that are too long are prone to non-specific hybridization, breakage, and other uncontrollable errors in the DNA storage process. To completely store the data in the DNA sequence, after the corresponding base sequence is obtained, it is divided into sequence blocks of equal length in DNA storage. This part of the data is called the payload (data bits). In addition to the stored data bits, appropriate non-payload bits are added, such as other redundant information embedded outside the stored sequence, including the primer, address bits, and parity bits. The higher the redundancy is, the more error-tolerant the final stored procedure is.

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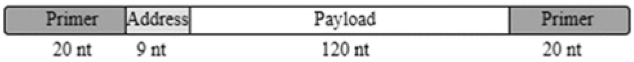

Fig. 1. The structure of a DNA storage sequence.

Considering the needs of sequencing and synthesis, we divide the data into sequence blocks with a length of 120 nt and set the address bits as sequence blocks with a length of 9 nt, as shown in Fig. 1. An address bit of length 9 can represent 29 different locations, which is a suitable size for a normal-size file. A. Encoding Method and Process of RSTE

To achieve a high encoding rate and encode more stable sequences, RTSE is pro-posed in this paper. The specific encoding process is as follows:

Step 1: Convert the file to be encoded into the corresponding 8-bit binary.

Step 2: If the text length of binary is longer than 500 bits, divide the final binary file into string segments with a length of 500 and then perform subsequent operations. If the text length is shorter than 500, directly find the longest repeating substring in the string through the LSBM algorithm and remaining substrings.

Step 3: Compare the repeated substring obtained in step 2 with the source binary, and find the frequency and position of the repeated substring from long to short. Store the substring, frequency, and position in an array.

Step 4: Generate map1 of the final string and frequency based on the Array in step 3.

Step 5: Generate the corresponding Huffman tree according to map1.

Step 6: Generate map2 of the substrings and corresponding base sequences ac-cording to the Huffman tree.

Step 7: Convert the source binary to the corresponding base sequence according to map2 generated in step 5 and map1 generated in step 3.Its main function is to splice the scattered data bits into the original long sequence. The front and back ends of the sequence are connected with primers of 20 nt length, which are used for PCR amplification and distinguishing data bits belonging to different files. In this way, only the relevant partial sequence.

can be read in the sequencing process, which can achieve random access and substantially reduce work repetition of sequencing and decoding. Regarding primers and address bits, Cao et al. [32], [33], and others have pro-posed a new encoding method for primers and address bits. The RSTE coding method in this paper is mainly aimed at data bits, and a sequence with a higher coding rate and more stable properties can be coded by this method. Primers are selected from the existing primer libraries [14].

Step 8: Attach address bits and primers to the payload.

To illustrate the encoding method proposed in this paper, as shown in Fig. 2, the text "One for all, all for one" is taken as an example to explain the coding method.

1) Huffman Tree: Sort the repeating substrings from longest to shortest to form a Huffman tree, and put the longest length at the front of the tree. Then, starting from the root node, which is the 0th level, the number of levels of each subsequent level of nodes is incremented by one. The left node of the odd-numbered level is coded as G, and the right node is coded as C. The left node of the even-numbered level is coded as A, and the right node is coded as T. Each repeating substring obtained by this encoding method can avoid two repeating bases at the same time, and the long base sequence splice can only have two consecutive bases at most. Incorporating the Huffman tree into the encoding method enables the encoded results to satisfy the required constraints of DNA storage sequences.

TABLE I
THE ENCODING RESULT OF THE CORRESPONDING STRINGS

S1	S2	S3	S4	S5	S6	S7	S8
TCT	ACA	ACT	TGA	TGT	AGT	AGA	TCA

Fig. 3 shows the result of constructing a binary tree for the aforesaid example text according to map1 in Section *B*. The numbers in Fig. 3 represent the number of levels. As shown in the figure, the left node of the odd level is coded as G, and the right node is coded as C. The left node of the even level is coded as A, and the right node is coded as T. The bottommost node is the corresponding substring. The final encoding results are shown in Table I.

2) Longest Substring Backtracking Method: In view of the shortcomings of the existing methods, this paper proposes a new encoding method. The main idea is to find the repeated substring in the source code and replace the longest repeated substring with the shortest encoding sequence, thereby improving the encoding efficiency and realizing lossless compression of data. In the process of converting the source code into the corresponding base sequence, the method that realizes this idea and can improve the coding rate the most is the LSBM.

The LSBM is a method used to find repeated substrings in a string. The classic method used to find repeated substrings in the past was brute-force matching, but this method has high time complexity and is prone to waste of hardware resources, resulting in low efficiency. According to the characteristics of binary arrangement—only two numbers and high repetition rate—we propose the LSBM. Its time complexity is O(n), and it can improve efficiency and reduce encoding time.

Suppose there are two strings S and L. The expressions of the two strings are shown in (1) and (2) below. Our goal is to find if there is overlap between S and L. If there is some same part P in L, as shown in (2), then in the next match, when the initial match does not hold, one can find the part in S that is the same as P, align the same part of L with it, and commence comparison, which saves the time of comparing S se-quentially from the next bit:

$$S_0, S_1, S_2, \dots S_m L_0, L_1, L_2, \dots L_n, n < m. \tag{1}$$

$$P_0, \dots, P_i = L_j, \dots L_{j+i} = L_k, \dots, L_{k+i}(0 \leq j, k \leq n-i, j \neq k). \tag{2}$$

Fig. 4 shows the process of comparing two strings. The repeated part P in string L is 11; hence, when comparing S and L, start from the beginning and align the part in S that is the same as P and the part in L that is the same as P for comparison. By this way, the repeated part only needs to be compared twice to determine whether there is a part of S that is exactly the same as a part of L. To judge the position where S and L start to compare

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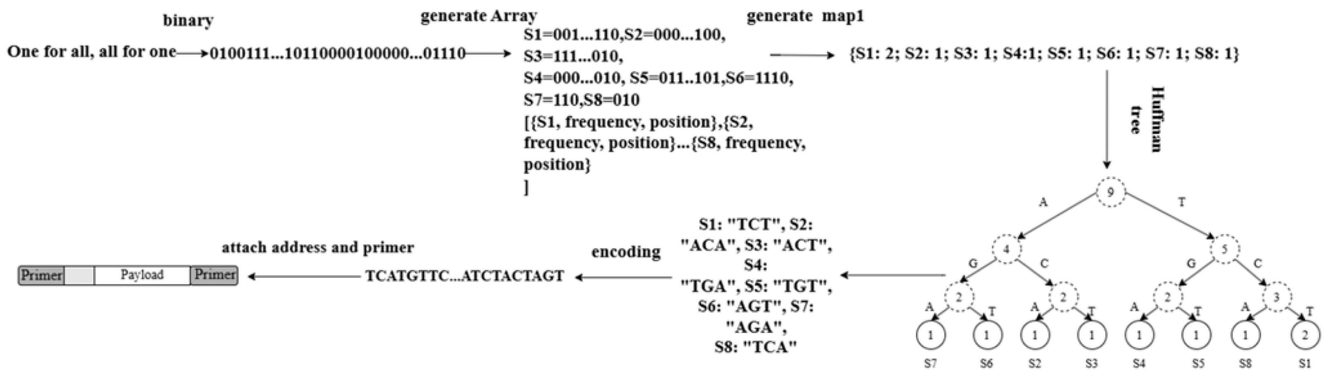

Fig. 2. Example of encoding process.

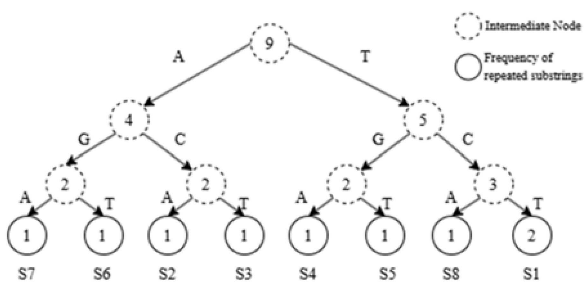

Fig. 3. Generating the Huffman tree according to map1.

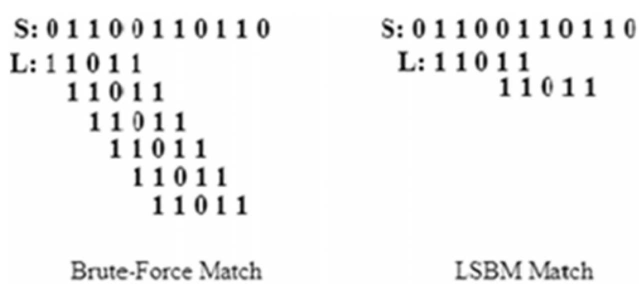

Fig. 4. The comparison process of brute-force match and LSBM Match.

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Pseudo Code of Longest Substring Backtracking Method
INPUT: STRING S, the prefix suffix substring L of S in order from longest to shortest
OUTPUT: are there parts in S that same as L and their locations
m←S.length, n←L.length, A is the array obtiained from the comparison algorithm, j←0
FOR i←0 to m
 DO IF S.i==L.j
    THEN i++,j++
    END IF
    IF j=n
    THEN S and L are identical
    END IF
    RETURN location is i-j    //S and L match, the position is i-j
    IF S.i!=L.j
    THEN IF j>0
        THEN j=A.j-1
        ELSE i++
        END IF
    END IF
 END FOR
 RETURN 0    //S and L ,not match
```

Fig. 5. Pseudo-code of the LSBM.

next time, we use array A to keep track of where each comparison should be compared. Fig. 5 explains the whole process of finding repeated substrings by the LSBM. The specific implementation logic of the LSBM is as follows:

Step 1: Input a binary string S whose length is less than or equal to 500.

Step 2: Get the suffix and prefix strings in S, and arrange them in order from long to short.

Step 3: Enter each prefix and suffix string in S in turn from long to short.

Step 4: Judge whether there is a part in S that is the same as the string according to the comparison array obtained for each string.

Step 5: Retain the suffix and prefix strings and positions existing in S, and form the final corresponding array.

B. Satisfied Constraints

When DNA sequences used for storage are stored in solution in single or double stranded form, the sequences with small differences in similarity are prone to non-specific hybridization. At the same time, the characteristics of DNA sequences also require the necessary biological properties to be taken into account during the coding process. Therefore, DNA mapping schemes must satisfy biological constraints to encode high-quality sequences suitable for storage [34]. This part of the paper introduces several related constraints and their functions. The sequences constructed by these constraints will be less error-prone during the sequencing and decoding processes [35].

1) Run-Length Limit Constraint: Continuous bases could lead to unstable molecular structures, unmanageable hybridization reactions of the whole sequence, and errors in decoding the continuous bases. Reading long homopolymers, such as the sequence ATGGGGTATG with four consecutive same bases, is especially prone to deletion errors. Therefore, during encoding, we apply the RLL constraint to try to avoid similar errors [36], [37].

The sequence constructed by the Huffman tree encodes substrings according to the positions of the left and right subtrees, which can effectively avoid the occurrence of two consecutive repeated bases. As the substrings have no repeating bases, the final coding result composed of the substrings according to positional encoding will only have two consecutive identical substrings at most; hence, the sequence will satisfy the RLL constraint.

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2) GC-Content Constraint: After generating the map table of the corresponding repeated substrings and positions by the LSBM, the GC content of the encoded sequence is guaranteed by Huffman tree encoding. In the Huffman encoding process, each substring of the source binary is encoded by the left and right subtrees of each Huffman tree. The nodes at odd levels are encoded as G and C, and the nodes at even levels are encoded as A and T. As shown in (3), for each substring, the difference between the number of GC bases and the number of AT bases is at most 1. This encoding method can maintain the GC con-tent of substring encoding at approximately 50%, and the GC content of the final en-coding result obtained through substring encoding according to positional encoding will also be maintained at approximately 50%:

$$n_{AT} - 1 \leq n_{GC} \leq n_{AT} + 1. \tag{3}$$

Since there are three hydrogen bonds in a GC base pair and two hydrogen bonds in an AT base pair, the content of G and C bases in a sequence has a large effect on the stability of the entire chain. If there is a certain proportion of GC content, the physical properties of the entire sequence can be more stable [13]. In DNA storage, the DNA strands with stable performance and stable structures are more suitable for storing data because that can reduce errors such as deletion and replacement of data.

3) End-Constraint: The End-constraint means that the last five bases of a short sequence cannot have more than three G and C bases [38]. This constraint can improve the stability of a sequence. The three constraints, namely, the RLL, GC-content, and end constraints, should be considered when converting the original data into base sequences. Since short sequences satisfy these constraints, long sequences composed of such short sequences also satisfy these constraints. Sequences that satisfy these constraints can be more stable in the actual DNA storage process, thus helping to reduce the occurrence of sequence breaks, secondary structures, and non-specific hybridization.

III. Comparions Bwtween RSTE and Other Encoding Methods

A. Encoding Efficiency

The coding rate is the most important factor in the encoding process. The theoret-ical storage limit of the quaternary model is 2 bits/nt, which is the limit of the storage efficiency of base sequences. Previous work has mainly focused on the goal of achiev-ing the theoretical maximum encoding rate; hence, the sequences used for comparison in this paper are the coding results under the guidance of the quaternary encoding method. As shown in Table IV, the binary data 00, 01, 10, and 11 are encoded as T, C, G, and A, respectively. Recent research has focused on improving the storage density, and the quaternary encoding method is the most widely used DNA storage encoding method [16], [25].

To better illustrate the effectiveness of the encoding method RSTE, this paper selects four different types of files and lists the encoding rate and performance results to substantiate the effectiveness and practicability. The four kinds of files are the stray bird of size 244 KB and file type of txt; the lyrics of "The Internationale" of size 684 KB and file type of pdf; the novel collection of O'Henry of size 764 KB and file type of txt; and a sunflower painting by Van Gogh of size 4.31 MB and file type jpg. In the subse-quent description, these four files are referred to as Txt1, Pdf, Txt2, and Jpg. The effec-tiveness of encoding can be tested more comprehensively by encoding files of different types and sizes. The final coding efficiency results are shown in Tables II, III, and IV, and the best results are shown in bold.

The superscript R indicates that the result is obtained by the RSTE encoding method proposed in this paper, and the superscript Q represents that the result is obtained by the quaternary encoding method. As shown in Table II, the size of the four files after encoding is reduced to 43.4%, 40.8%, 44.8%, and 62.2% of the original file, which proves that DNA storage has a high information density compared to traditional binary-based hardware storage. Compared with the most commonly used quaternary encoding method, the encoding result size of RSTE is reduced by 13.1%, 18.4%, 10.5%, and 10%. It can be seen from the data in the table that the RSTE coding method can substantially improve the coding rate, and the coding rate is increased on the basis of the theoretical maximum coding rate. The theoretical maximum encoding rate is 2. Using RSTE, the encoding rates of files improved to varying degrees. The file encoding rate of pdf had the highest rate of improvement, which was 22.5%. The average encoding rate for the four files was 2.26, which is a 13% improvement over the theoretical maximum.Compared with previous methods, the encoding method proposed in this work substantially improves the encoding rate. Higher coding rates mean shorter sequences, which can substantially reduce the cost of DNA storage, and shorter sequences also mean higher synthesis and sequencing efficiencies. Table II shows the encoding results and net information density of different files with the encoding method proposed in this paper. "Data bits" are the number of bases in data bits; "Bases" represents the number of bases added with primers and address bits after encoding; "Sequence number" represents the number of encoded sequences; and "Bases (no primer)" represents the bases that remove primers after encoding.

To prove that the RSTE encoding method has better results compared with other methods for a wide range of applications, we compared RSTE with several encoding methods in terms of different factors. The results are shown in Table IV. In the table, "Data size" refers to the file size. "Encoding rate" refers to the average number of bits in the synthesized sequence. "Redundancy" refers to the coverage of the sequence (for example, in the method by Goldman et al., the information from one sequence is also encoded into three additional sequences). "Full recovery" refers to whether the original data can be completely recovered. "NID" represents Net information density (bits/nt), with its calculation detailed in Formula (4). It can be seen from the data in the table that the method proposed in this paper improved all indicators compared with previous work. The coding rate increased by 16%–126%. In addition, the encoding method has minimal coverage and can also achieve lossless conversion of the sequence to the original binary.

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TABLE II
COMPARISON OF RSTE AND THE QUATERNARY ENCODING METHOD

Scheme	Txt1	Pdf	Txt2	Jpg
	244	684	764	4413
Original file size (KB)	122^{Q}	342^{Q}	382^{Q}	2570^{Q}
	106^{R}	**279^{R}**	**342^{R}**	**1986^{R}**
Encoding rate	2^{Q}	2^{Q}	2^{Q}	2^{Q}
	2.31^{R}	**2.45^{R}**	**2.24^{R}**	**2.03^{R}**
Bases	249506^{Q}	699519^{Q}	782184^{Q}	4525208^{Q}
	108046^{R}	**285468^{R}**	**349394^{R}**	**222979^{R}**

TABLE III
NET INFORMATION DENSITY OF RSTE

File	Data bits	Bases	Sequence number	Bases (no primer)	Net information density (bits/nt)
Txt1	108046	152195	901	116164	2.15
Pdf	285468	402039	2369	306879	2.28
Txt2	349394	492082	2912	375602	2.08
Jpg	2229799	3140317	18582	2397037	1.89

TABLE IV
COMPARISON WITH DIFFERENT ENCODING MATHODS

	Church[2]	Goldman[12]	Grass[13]	Bornholt[14]	Organick[18]	Ping[26]	Welzel[27]	RSTE
Data size(MB)	0.63	0.73	0.08	0.15	200.2 / 0.03	20.2	0.005	**5.96**
Encoding rate(bits/nt)	1	1.58	1.78	1.58	1.45	1.95	/	**2.26**
Redundancy	1	4	1	1.5	1.15	1	1	**1**
Full recovery	No	No	Yes	No	Yes	Yes	Yes	**Yes**
NID（bits/nt）	0.83	0.33	1.14	0.8	1.1	1.25	1.17	**2.1**

Net information density is calculated by dividing the number of bits of input information by the number of synthesized DNA bases (no primers), as shown in (4):

$$Net\ Information\ Density = \frac{Bits\ of\ input\ information}{Synthesized\ DNA\ Bases\ (no\ primer)} \quad (4)$$

B. Encoding Quality

In addition to the encoding rate, the encoding quality is also an important factor to assess a coding method. The encoding quality is a more important factor than the encoding rate. DNA sequences as storage media have completely different physical and thermodynamic properties for hardware storage. Sequences in solution will have special situations such as non-specific hybridization, secondary structures, and breakage, which should be avoided as much as possible by the choice of the coding process. DNA encoding is the beginning of the DNA storage process. It is necessary to encode high-quality storage sequences to avoid errors in subsequent steps, and it is also necessary to increase the encoding rate to reduce the storage cost. Therefore, this paper first evaluates the stability of the encoded sequences from a macroscopic perspective by constraints on GC content, run-length, and end constrain as part of the proposed RSTE encoding method. These constraints serve to control overall sequence properties that influence synthesis and sequencing reliability. Subsequently, the stability of the resulting sequences is assessed from a microscopic perspective, using quantitative and qualitative analyses based on minimum free energy (MFE) and melting temperature (Tm).

1) Comparison of Constrains: In DNA storage technology, evaluating encoding quality through biochemical constraints—such as GC content regulation, homopolymer limitations, and terminal sequence constraints—serves as a cornerstone strategy for achieving efficient and reliable storage. These constraints not only establish well-defined biochemical boundaries for encoding design but also significantly enhance the comprehensive performance of DNA storage systems through multidimensional optimization. As shown in Table V, this paper compares with previous work under four different constraints. The specific analysis is as follows:

Run-length Limit constraint systematically mitigate errors inherent in synthesis and sequencing processes. This paper adopts a run-length constraint of 1, as shown in Table V. In comparison, most existing encoding schemes impose more relaxed constraints, typically limiting the maximum run length to 3 or 4. It is worth noting that both the method proposed in

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TABLE V
COMPARISON OF DIFFERENT CONSTRAINS

Methods	GC Content (%)	Run-length Limit	End-constraint
Church[2]	2.5~100	3	No
Goldman[12]	22.5~82.5	1	No
Grass[13]	12.5~100	3	No
Erlich[20]	40~60	4	No
Ping[26]	40~60	4	No
Welzel[27]	40~60	3	No
RSTE	**40~50**	**1**	**Yes**

TABLE VI
COMPARISON OF GC CONTENT BETWEEN RSTE AND THE QUATERNARY ENCODING METHOD

Scheme	Quaternary	RSTE
Txt1	47.4%	**42.8%**
Pdf	35.6%	**43.1%**
Txt2	59.1%	**43.2%**
Jpg	38.9%	**46.4%**
Mean	45.25%	**43.9%**
Variance	0.09	**0.00021**

this paper and the approach by Goldman et al. enforce a stricter run-length constraint, limiting consecutive identical nucleotides to a maximum of 1.

Furthermore, the end-constraint helps eliminate undesirable secondary structures such as hairpin loops, thereby concentrating the free energy distribution and reducing fluctuations in melting temperature (Tm). This, in turn, significantly enhances the uniformity and reliability of large-scale parallel synthesis and sequencing. However, this constraint is not implemented in the other schemes. A detailed analysis of the free energy concentration achieved by the proposed method is presented in the following section.

Precise control of GC content (guanine-cytosine ratio, typically set at 40%–60%) balances the thermodynamic stability of DNA molecules. As shown in Table V, among all the compared methods, only the proposed scheme, along with the approaches by Erlich et al., Ping et al., and Welzel et al., strictly constrain the GC content within the range of 40% to 60%. The remaining schemes do not limit GC content constraints. As the most important evaluation index in the DNA storage encoding method, this paper also proves the stability of the sequence by comparing the variance of the GC content of each sequence, as shown in Table VI, “Quaternary” refers to the quaternary encoding method, and “RSTE” refers to the encoding method proposed in this paper. In the coding results with our method, the GC content is maintained between 40% and 50%, which satisfies the GC-content constraint. However, the GC content of sequences under the quaternary encoding method is quite different, ranging from 30% to 60%, which does not meet the GC-content constraint. The variance of the GC content of the four files is 0.09 under the quaternary encoding method, whereas it is 0.00021 under the encoding method proposed in this paper. It can be seen that RSTE can construct a sequence with more

TABLE VII
COMPARISON OF T_M BETWEEN RSTE AND THE QUATERNARY ENCODING METHOD

Scheme	Txt1	Pdf	Txt2	Jpg
Average F_{Tm}	3.99 [Q]	14.19 [Q]	3.24 [Q]	5.34 [Q]
	0.53 [R]	**0.44 [R]**	**0.49 [R]**	**0.28 [R]**
Average T_m	98.25 [Q]	85.97 [Q]	98.89 [Q]	98.16 [Q]
	90.29 [R]	**90.41 [R]**	**90.46 [R]**	**91.81 [R]**

TABLE VIII
COMPARISON OF $\delta(G)$ BETWEEN QUATERNARY AND RSTE ENCODING

δ(G)	Txt1	Pdf	Txt2	Jpg
Quaternary	45.34	126.43	31.65	50.17
RSTE	**7.36**	**5.97**	**6.89**	**4.52**

stable GC content, and the constructed sequences are also more suitable for DNA storage.

2) Comparison of Melting Temperature: The Tm value is used to measure the stability of the melting temperature of a sequence candidate set. The mean value of the melting temperature can indicate the stability of the Tm value in a sequence set [39]. If the differences in the melting temperatures between each sequence are big, the candidate set has an unstable state. FTm(S) represents the difference in Tm values:

$$F_{Tm}(S) = \sum_{j=1}^{n} \left\{ Tm(S_j) - \overline{Tm}(S) \right\}^2 \tag{5}$$

where $\overline{Tm}(S)$represents the mean melting temperature of *S*, and $Tm(S_j)$represents the melting temperature of the *jth* sequence in candidate set *S*.

Table VII shows the comparison of the FTm and mean melting temperatures between the quaternary and RSTE encoding methods. The smaller the FTm value is, the smaller the difference in the melting temperature between sequences is, and the more stable the thermodynamic property of the sequence. In DNA storage, due to the uncertainty of non-specific hybridization and mutation of sequences, sequences with a stable thermodynamic property are more suitable to avoid errors during storage and decoding [40]. It can be seen from the results in Table VII that the stability of the four files is increased by 7.5 times, 32 times, 6.6 times, and 19 times, and the difference be-tween the melting temperatures of the same file is substantially reduced. The stability of Tm of the pdf is improved to the greatest extent. It can be seen that the proposed en-coding method can construct a more stable data bit storage sequence. The average melting temperature of the sequences encoded by the quaternary encoding method is maintained at 85.97°C–98.89°C, whereas the melting temperature of the sequences encoded by the RSTE method is 90.29°C–91.81°C. By using the encoding method pro-posed in this paper for DNA storage, it is possible to obtain data bit sequences with approximately the same melting temperature even for files of different types and sizes. Therefore, when the synthesis and sequencing work is performed, the data bit se-quences of different files can be placed in the same wet room for operation, thereby reducing the complexity of the work.

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The DNA storage encoding method proposed in this paper can encode sequences with more stable melting temperature and also maintain the melting temperature difference in a smaller range.

3) Comparison of Minimum Free Energy: The Minimum Free Energy (MFE) is an parameter to measures a hybridization re-action [41]. It refers to the energy released by the hybridization reaction. When DNA storage synthesis is completed, the information is stored as a double strand, and two sequences have to release energy to form a base pair structure. That is, the MFE must be less than 0 for the double-strand structure to be stable. The MFE in this work is calculated by PairFold [41].

To indicate the difference in the MFE between sequences, we use $\delta(G)$. $\delta(G)$ is the variance of the MFE, and it represents the stability of the MFE in the whole set. Larger values of (G) indicate larger differences in the MFE between sequences. Table VIII shows the MFE difference between data bits under the two different encoding methods. It can be seen from the table that the diffrences in the MFE are reduced by 83.8%, 95.3%, 78.2%, and 91%, and the stability of the MFE between sequences is substantially improved. This shows that the data bits encoded with the encoding method proposed in this paper have more stable thermodynamic properties.

IV. Conclusion

This paper proposes an encoding method named the RSTE Method. Its encoding process is as follows. First, convert the data or file into binary, and then, generate the corresponding hash table through the LSBM algorithm. Next, generate the Huffman tree according to the hash table to generate the encoding result. Finally, produce the final encoding result under the position and encoding results. To demonstrate the efficiency of RSTE, we used the encoding method to encode four different types and sizes of files. By comparing the encoding rates with other work, we found that this method can increase the theoretical maximum encoding rate by a maximum of 22.5%. The average encoding rate of the four files was 2.26, which is 13% higher than the theoretical maximum. Since biological constraints reduce the actual encoding rate of DNA information storage, some encoding methods sacrifice the encoding rate to meet the biological constraints. Different from the previous studies on encoding methods, this paper pays more attention to constraints that the sequence should satisfy in the actual storage process while improving the encoding rate. The encoded sequences by RSTE satisfy the three biological constraints: RLL, GC-content, and end constraints.

In addition, the sequences encoded by this encoding method also have more stable properties. We measure the stability of the sequence by three different indicators: control of different constraints, melting temperature, and minimum free energy. For the control of biological constraints, the RSTE method can control the three constraints to the optimal state, proving the stability of the encoded sequence. For melting temperature, we used the $\mathrm{F_{Tm}}$ to compare the difference in the melting temperature between different sequences and used the average melting temperature to compare the melting temperatures of sequences. Through experimental results, we found that the sequence encoded by the same file has a more stable melting temperature under our encoding method, and the difference in the melting temperature of different files can also be controlled in a smaller range. The stability of the MFE between sequences encoded by different types of files was improved by 78.2%–91%, and the gap was greatly reduced. The DNA storage encoding method proposed in this paper can encode sequences with more stable thermodynamic properties, thereby improving the efficiency of synthesis and sequencing in DNA storage and enhancing the convenience of operations.

Although our encoding method substantially improves the encoding rate, the encoding time is too long due to the constant back-and-forth comparison required to find the longest identical substring. In addition, the encoding rates of different types of files are quite different, which will lead to a lower encoding rate of this method for certain files [42]. In future work, we will pay more attention to reducing the time complexity of the encoding method and improving the time efficiency of encoding, and applying the high density of DNA storage to medical images [43], [44], cultural relics protection and other fields.

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